

Assignment-4:Optimizing Transformer Translation with RayTune and Optuna

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1 Objective

The objective of this assignment is to build a Transformer-based Neural Machine Translation (NMT) model from scratch using PyTorch to translate English sentences to Hindi. Furthermore, the assignment involves optimizing the model's hyperparameters using **Ray Tune**, **Optuna**, and the **ASHA Scheduler** to maximize the BLEU score.

2 Dataset and Preprocessing

- **Data:** An English-Hindi TSV dataset comprising 13,186 sentence pairs.
- **Vocabulary:** Custom vocabulary built with a minimum frequency threshold of 2.
- **Padding:** Sentences padded/truncated to a maximum length of 50 tokens.

3 Part 1: Baseline Model

A standard Encoder-Decoder Transformer model was implemented with the following default configuration:

- **Architecture:** $d_{model} = 512$, $N_{layers} = 6$, $N_{heads} = 8$, $d_{ff} = 2048$, Dropout = 0.1.
- **Training:** Batch size = 60, Learning Rate = 10^{-4} , Epochs = 100.
- **Performance:** The baseline model achieved a **NLTK BLEU Score of 49.71**.

4 Parts 2 & 3: Hyperparameter Optimization

To improve the baseline model, hyperparameter tuning was conducted using **Ray Tune** combined with **OptunaSearch** (for efficient search space exploration) and **ASHAScheduler** (for aggressive early stopping of poorly performing trials).

4.1 Search Space

The following hyperparameter search space was defined:

- **Learning Rate:** LogUniform (10^{-5} to 10^{-3})
- **Batch Size:** Categorical {16, 32, 64}
- **Attention Heads (N_{heads}):** Categorical {4, 8}
- **FeedForward Dim (d_{ff}):** Categorical {1024, 2048}
- **Dropout:** Uniform (0.1 to 0.4)

4.2 Best Configuration Found

The Optuna search yielded the following optimal configuration (minimizing Cross-Entropy Loss):

- **Learning Rate:** 1.107×10^{-4}
- **Batch Size:** 32
- **Attention Heads:** 4
- **FeedForward Dim:** 1024
- **Dropout:** 0.1827

5 Final Training and Results

The Transformer model was re-trained from scratch using the best hyperparameters discovered by Ray Tune for 90 epochs.

5.1 Performance Comparison

The hyperparameter optimization yielded a massive improvement in translation quality.

Model	Epochs	BLEU Score
Baseline Transformer	100	49.71
Tuned Transformer	90	73.81

6 Conclusion

Implementing Ray Tune with Optuna and the ASHA Scheduler successfully identified a leaner, more regularized architecture (fewer attention heads, smaller feedforward dimension, and higher dropout) coupled with an optimal learning rate and batch size. This optimized configuration boosted the evaluation BLEU score from **49.71** to **73.81**, demonstrating the critical importance of systematic hyperparameter tuning in NMT workflows.